Telegram AI Booking System

Project Overview

This project implements an automated AI-powered service booking system using Telegram as the customer interface. Customers chat with an intelligent bot to browse available services, ask questions, and complete bookings. All booking data is automatically saved to Google Sheets and a confirmation email is sent via Gmail. The AI is powered by Groq's free LLM API.

Tech Stack

Tool	Purpose	Cost
n8n (Self-hosted)	Workflow automation engine	Free
Telegram Bot API	Customer chat interface	Free
Groq AI (llama-3.3-70b-versatile)	AI conversation and booking logic	Free
Google Sheets	Products and bookings database	Free
Gmail	Confirmation email sender	Free
ngrok	Expose local n8n to internet	Free

Features

- AI-powered natural conversation via Telegram
- Auto product/service listing fetched from Google Sheets
- Booking collection - Name, Email, Date, Time
- Auto-save confirmed bookings to Google Sheets
- Automatic HTML confirmation email sent to customer
- Unique Booking ID generated for each booking

System Architecture - Data Flow

Customer Message (Telegram)

↓

Telegram Trigger (n8n)

```
    ↓
Extract Message Data
    ↓
Fetch Products from Google Sheets
    ↓
Prepare AI Context
    ↓
Groq AI Response (llama-3.3-70b-versatile)
    ↓
Parse AI Response
    ├── Send Reply ──→ Telegram (customer sees message)
    └── Is Booking Confirmed?
        ↓ true
    Save Booking to Google Sheets
        ↓
    Send Confirmation Email (Gmail)
```

Prerequisites

- Node.js v18 or higher - nodejs.org
- n8n installed globally
- Telegram account
- Google account (for Sheets and Gmail)
- Groq account - console.groq.com (free)
- ngrok account - ngrok.com (free)

Setup Instructions

Step 1 - Install and Start n8n

npm install -g n8n

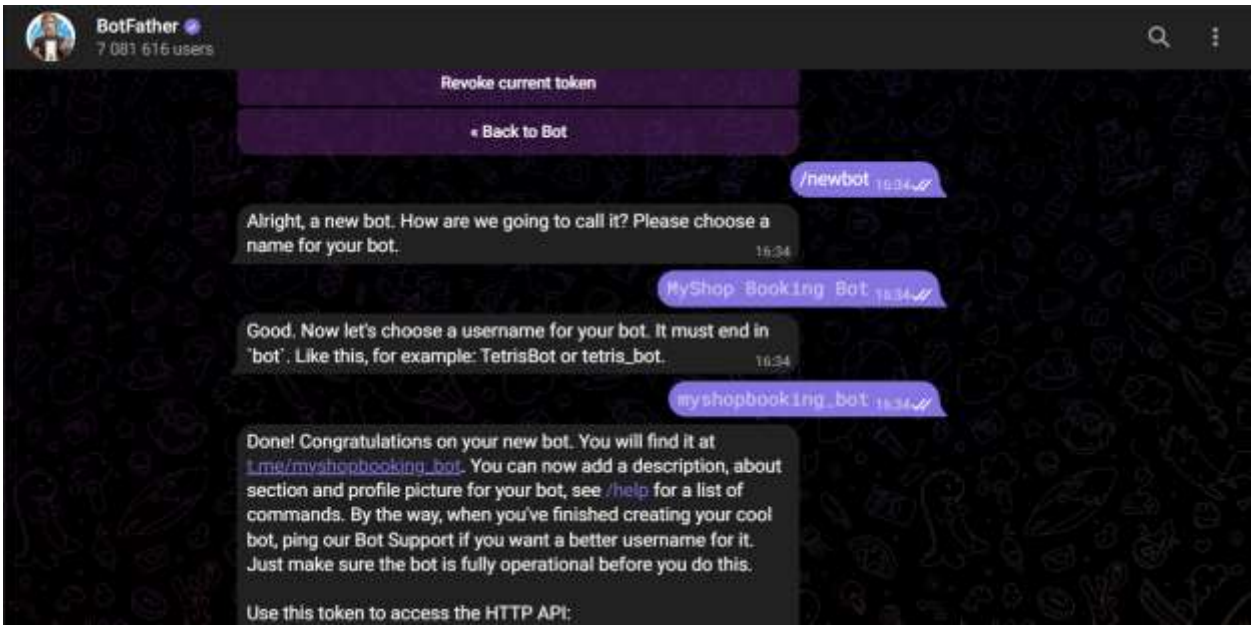
set WEBHOOK_URL=https://your-ngrok-url.ngrok-free.app/

n8n start

Open browser at: http://localhost:5678

Step 2 - Create Telegram Bot

1. Open Telegram and search for @BotFather
2. Send: /newbot
3. Enter a display name: MyShop Booking Bot
4. Enter a username ending in bot: myshopbooking_bot
5. Copy the API Token provided by BotFather
6. In n8n: Credentials → Add → Telegram API → paste token → Save



Step 3 - Get Groq API Key (Free)

1. Go to console.groq.com and sign up with Google (free, no credit card)
2. Navigate to API Keys in the left menu
3. Click Create API Key and copy it
4. In n8n HTTP Request node, set Authorization header to: Bearer gsk_your_key_here

5. Model used: llama-3.3-70b-versatile

Step 4 - Google Sheets Setup

Create a Google Spreadsheet with two sheets:

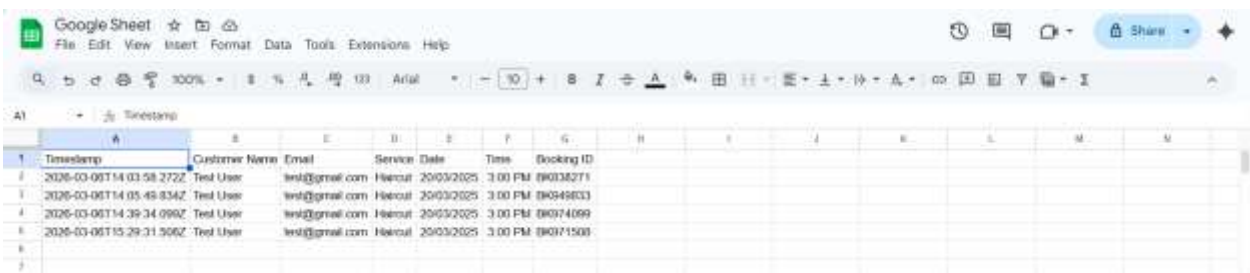
Sheet 1 - Products:

Product Name	Description	Price	Availability	Duration	Category
Haircut	Basic haircut service	500	Yes	30 mins	Grooming
Facial	Deep cleansing facial	1200	Yes	60 mins	Skincare

Sheet 2 - Bookings (auto-filled by system):

Timestamp	Customer Name	Email	Service	Date	Time	Booking ID

Enable Google Sheets API and Google Drive API from Google Cloud Console. Create OAuth 2.0 credentials (Desktop app type). Add the credentials in n8n: Credentials → Google Sheets OAuth2 API.



Step 5 - Import Workflow in n8n

1. Open http://localhost:5678
2. Go to Workflows → Add Workflow
3. Click the 3-dot menu (⋮) at top right
4. Select Import from JSON
5. Paste the contents of workflow.json and click Import

Step 6 - Configure Each Node

Node	What to Configure
Telegram Trigger	Select Telegram Bot credential

Node	What to Configure
Get Products from Sheet	Select Google Sheets credential + enter Sheet ID
HTTP Request (Groq)	Authorization header: Bearer gsk_your_key, model: llama-3.3-70b-versatile
IF Node	Condition: Boolean, $json.hasBooking is true
Save Booking to Sheet	Google Sheets credential + Sheet ID, Operation: Append Row
Send Confirmation Email	Gmail OAuth2 credential

Step 7 - Start ngrok and Set Webhook

Terminal 1 - already running n8n

Terminal 2:

ngrok http 5678

Copy the https URL from ngrok output, then open this in browser:

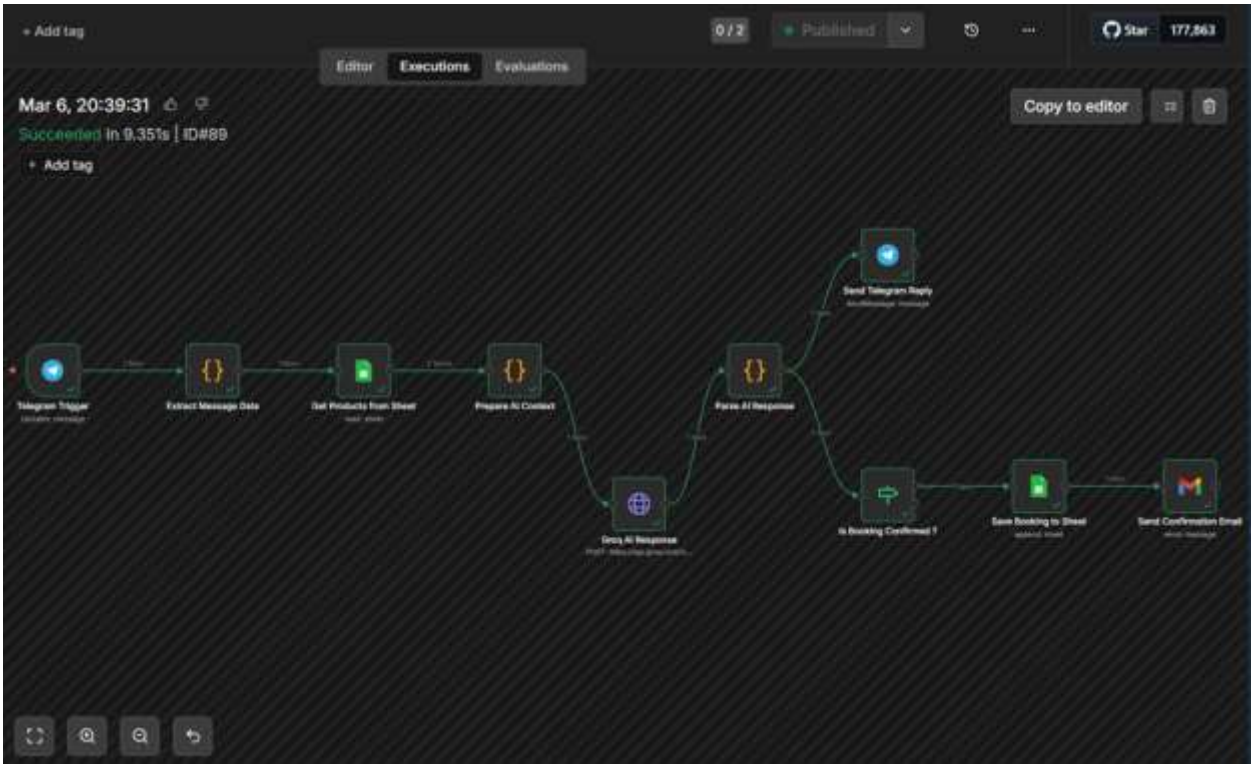
https://api.telegram.org/bot8237444554:AAHID2zivJVwDbI02HvIZdX9YXV7_UCWGnw/setWebhook?url=https://eloise-spectaclelike-kenda.ngrok-free.dev/webhook/telegram-webhook-001

You should see: {"ok":true,"result":true}

```
C:\Windows\System32\cmd.e
ngrok                                                                    (Ctrl+C to quit)

*  One gateway for every AI model. Available in early access *ngrok*: https://ngrok.com/r/ai

Session Status                online
Account                       smdeep137@gmail.com (Plan: Free)
Version                       3.37.0
Region                        India (in)
Latency                       39ms
Web Interface                 http://127.0.0.1:4040
Forwarding                    https://eloise-spectaclelike-kenda.ngrok-free.dev -> http://localhost:5678

Connections                   ttl     opn     rt1     rt5     p50     p90
                              14032   0       0.00    0.00    0.61    6.23

HTTP Requests
-------------

21:29:22.799 +26 POST /webhook/telegram-webhook-001/webhook 200 OK
21:17:23.827 +26 POST /rest/ph/i/v0/e/                      204 No Content
21:09:42.174 +26 GET  /rest/executions                      304 Not Modified
21:09:37.106 +26 GET  /rest/executions                      304 Not Modified
21:09:32.054 +26 GET  /rest/executions                      304 Not Modified
21:09:26.982 +26 GET  /rest/executions                      304 Not Modified
21:09:21.929 +26 GET  /rest/executions                      304 Not Modified
21:09:16.279 +26 GET  /healthz                              200 OK
21:09:16.813 +26 GET  /rest/executions                      304 Not Modified
21:09:16.305 +26 GET  /healthz                              200 OK
```

Step 8 - Activate Workflow

In n8n, open the workflow and toggle the Active switch to ON. The system is now live.



Usage - Example Conversation

User: Hello

Bot: Hi! Welcome to our booking system!

Here are our available services:

1. Haircut - Basic haircut | Price: 500 BDT | Duration: 30 mins

2. Facial - Deep cleansing | Price: 1200 BDT | Duration: 60 mins

User: I want to book a haircut

Bot: Great! Please provide:

- Your full name

- Email address

- Preferred date (DD/MM/YYYY)

- Preferred time

User: Rahman Ali, rahman@gmail.com, 20/03/2025, 3 PM

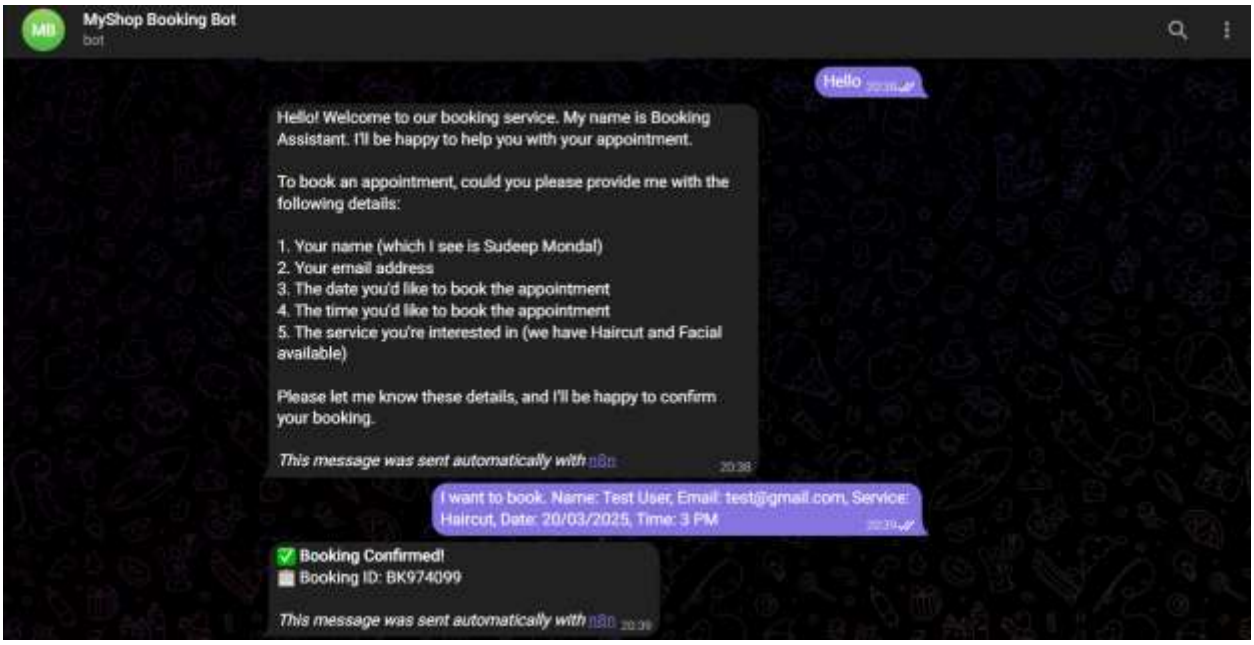
Bot: ✅ Booking Confirmed!

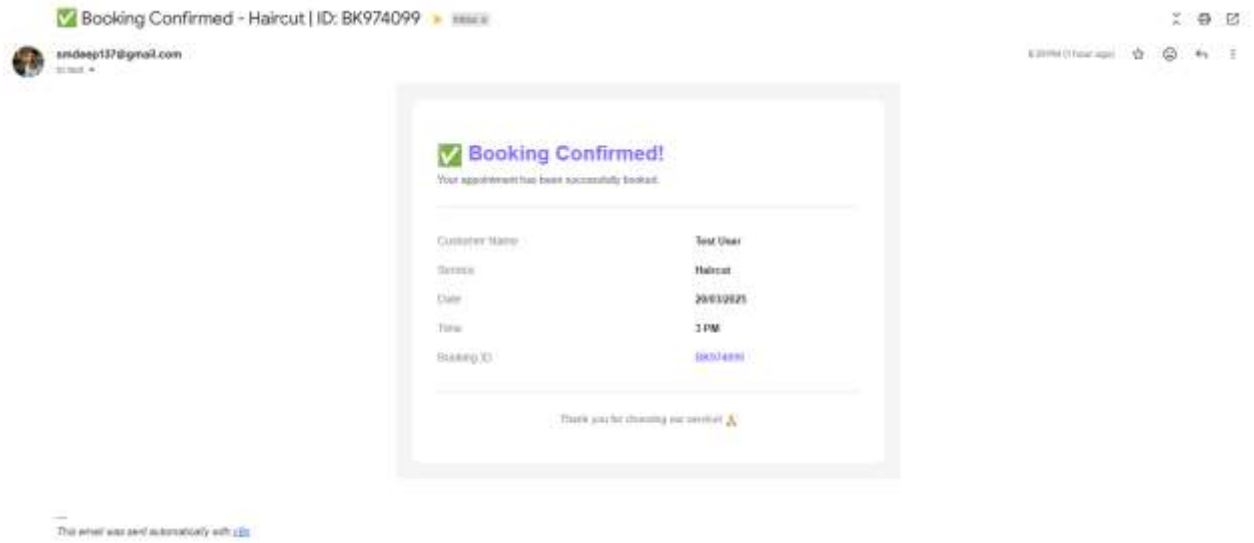
📋 Booking ID: BK492841

(Confirmation email sent to rahman@gmail.com)

After booking confirmation:

- Row automatically added to Google Sheets Bookings tab
- HTML confirmation email sent to customer
- Unique Booking ID generated (format: BK + 6 digits)





Workflow Node Details

Telegram Trigger - Listens for incoming messages via webhook. Receives all message types from customers.

Extract Message Data - Code node that extracts text, chatId, firstName, and userId from the raw Telegram payload.

Get Products from Sheet - Reads all rows from the Products sheet to get current service listings dynamically.

Prepare AI Context - Formats product data into a readable list and combines it with the user message to build AI context.

HTTP Request (Groq AI) - Sends a POST request to Groq API with the system prompt and user message. Returns AI-generated response.

Parse AI Response - Parses the AI text, detects the BOOKING_CONFIRMED block using regex, and extracts all booking fields.

Send Telegram Reply - Sends the AI response back to the customer in Telegram with Markdown formatting.

IF Node - Checks if hasBooking is true. Routes to the booking save path only when a booking is confirmed.

Save Booking to Sheet - Appends a new row to the Bookings sheet with all booking details including timestamp and booking ID.

Send Confirmation Email - Sends an HTML-formatted confirmation email to the customer's email address.

Error Handling

Error	**Cause**	**Solution**
Bot not responding	Webhook not set or workflow inactive	Activate workflow, re-run setWebhook with current ngrok URL
Groq model error	Decommissioned model name	Use: llama-3.3-70b-versatile
Sheet not saving	IF condition wrong or operation not Append	Set IF to Boolean equals true, Sheet operation to Append Row
Email not sending	Gmail credential not authorized	Re-connect Gmail OAuth2 in n8n Credentials
ngrok URL changed	ngrok restarted on free plan	Re-run setWebhook with new URL, restart n8n with new WEBHOOK_URL
Booking not detected	AI not following booking format	Check Parse AI Response output in n8n Executions tab

Project Structure

```
telegram-booking-bot/
├── workflow.json      ← n8n workflow export (import this)
├── .env.example       ← API keys template
├── .gitignore
└── README.md          ← This documentation
```

Environment Variables

TELEGRAM_BOT_TOKEN=8237444554:AAHID2zivJVwDbI02HvIZdX9YXV7_UCWGnw

GROQ_API_KEY=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

GOOGLE_SHEET_ID=https://docs.google.com/spreadsheets/d/1jZuNN0UFhM4jR6v9h4kQ51nD-dOOkUiQrehNLdzDuas/edit?gid=0#gid=0

PRODUCTS_SHEET_NAME=1jZuNN0UFhM4jR6v9h4kQ51nD-dOOkUiQrehNLdzDuas

BOOKINGS_SHEET_NAME=1jZuNN0UFhM4jR6v9h4kQ51nD-dOOkUiQrehNLdzDuas